
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Linear Algebra Revision 1

The following exercises revisit some concepts from linear algebra. These are more mathematically involved. I would encourage you to try them (especially Exercise 1). However, you can skip them for the time being and come back later.

Exercise 1. Here, we consider the general definition of a *vector space*. Let $\mathcal{V}$ be a collection of objects, equipped with two operations

$$+ : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V} \quad \text{and} \quad \cdot : \mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V}.$$

The collection $\mathcal{V}$ with $+$ and $\cdot$ is called a **vector space** if the following hold.

- (A1) $\mathbf{x}, \mathbf{y} \in \mathcal{V} \Rightarrow \mathbf{x} + \mathbf{y} \in \mathcal{V}$ ($\mathcal{V}$ is **closed** under $+$)
- (A2) $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x} \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ ($+$ is **commutative**)
- (A3) $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z} \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ ($+$ is **associative**)
- (A4) There exists $\mathbf{0} \in \mathcal{V}$ s.t. $\mathbf{x} + \mathbf{0} = \mathbf{x} \forall \mathbf{x} \in \mathcal{V}$. Such an element $\mathbf{0}$ is called the **additive identity**.
- (A5) For every $\mathbf{x} \in \mathcal{V}$, there exists a $\mathbf{y} \in \mathcal{V}$ such that $\mathbf{x} + \mathbf{y} = \mathbf{0}$. Such a $\mathbf{y}$ is called the **additive inverse** of $\mathbf{x}$ and is denoted by $-\mathbf{x}$.
- (M1) $\mathbf{x} \in \mathcal{V} \Rightarrow a \cdot \mathbf{x} \in \mathcal{V} \forall a \in \mathbb{R}$ ($\mathcal{V}$ is **closed** under $\cdot$)
- (M2) $a \cdot (b \cdot \mathbf{x}) = (ab) \cdot \mathbf{x} \forall a, b \in \mathbb{R}, \mathbf{x} \in \mathcal{V}$ ($\cdot$ is **commutative**)
- (M3) $(a + b) \cdot \mathbf{x} = a \cdot \mathbf{x} + b \cdot \mathbf{x} \forall a, b \in \mathbb{R}, \mathbf{x} \in \mathcal{V}$ ($+$ is **distributive** over $\cdot$)

Note. Here, the $+$ on the left (i.e., in $a + b$) is the usual addition (between two real numbers), whereas the $+$ on the right (i.e., in $a \cdot \mathbf{x} + b \cdot \mathbf{x}$) is the “binary” operation $+$ defined at the beginning of the exercise.

- (M4) $a \cdot (\mathbf{x} + \mathbf{y}) = a \cdot \mathbf{x} + a \cdot \mathbf{y} \forall a, b \in \mathbb{R}, \mathbf{x} \in \mathcal{V}$ ($\cdot$ is **distributive** over $+$).
- (M5) $1 \cdot \mathbf{x} = \mathbf{x} \forall \mathbf{x} \in \mathcal{V}$.

The elements of a vector space $\mathcal{V}$ (equipped with $+$ and $\cdot$) are called vectors. The operation $+$ is called **vector addition** and $\cdot$ is called **scalar multiplication**.

Check that the following are vector spaces:

- (a) $\mathbb{R}^n$ – n -tuples of real numbers $\mathbf{x} = (x_1, \dots, x_n)^\top$, where $x_1, \dots, x_n \in \mathbb{R}$. The operations $+$ and $\cdot$ are the usual addition and multiplication of real numbers used elementwise. That is, for $\mathbf{x} = (x_1, \dots, x_n)^\top$, $\mathbf{y} = (y_1, \dots, y_n)^\top$, $\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n)^\top$; and for $a \in \mathbb{R}$, $a \cdot \mathbf{x} = (ax_1, \dots, ax_n)^\top$.

Note: The general definition DOES NOT require vector addition or scalar multiplication to be applied elementwise. In fact, the general definition DOES NOT require vectors to be an ordered tuple of elements! See an example in part (d) below.

- (b) $\mathbb{C}^n$ – n -tuples of complex numbers $\mathbf{z} = (z_1, \dots, z_n)^\top$, where $z_1, \dots, z_n \in \mathbb{C}$ (i.e., $z_j = x_j + iy_j$). The operations $+$ and $\cdot$ are the usual addition and multiplication applied elementwise. That is, for $\mathbf{z} = (z_1, \dots, z_n)^\top$ and $\mathbf{w} = (w_1, \dots, w_n)^\top$, $\mathbf{z} + \mathbf{w} = (z_1 + w_1, \dots, z_n + w_n)^\top$; and for $a \in \mathbb{R}$, $a \cdot \mathbf{z} = (az_1, \dots, az_n)^\top$. For complex numbers $z = x + iy$ and $w = u + iv$, we define the addition and scalar multiplication usually as $z + w = (x + u) + i(y + v)$ and $az = (ax) + i(ay)$.
- (c) $\mathbb{R}^{m \times n}$ – real matrices of order $m \times n$: $A = ((a_{ij}))_{i=1, \dots, m, j=1, \dots, n}$ with $a_{ij} \in \mathbb{R}$. The operations $+$ and $\cdot$ are the usual addition and multiplication of real numbers applied elementwise. Check the same for $\mathbb{C}^{m \times n}$ – complex matrices of order $m \times n$: $C = ((c_{ij}))_{i=1, \dots, m, j=1, \dots, n}$ with $c_{ij} \in \mathbb{C}$.
- (d) The space of all functions $\mathcal{F} = \{f : \mathbb{R} \rightarrow \mathbb{R}\}$. The operations $+$ and $\cdot$ are defined as: $(f + g)(x) = f(x) + g(x) \forall x \in \mathbb{R}$; and for $a \in \mathbb{R}$, $(a \cdot f)(x) = af(x) \forall x \in \mathbb{R}$.
- (e) The space of all continuous functions $\mathcal{F}_c = \{f : \mathbb{R} \rightarrow \mathbb{R} \text{ continuous}\}$. The operations $+$ and $\cdot$ are defined as above.
- (f) The space of all integrable functions $\mathcal{F}_1 = \{f : \mathbb{R} \rightarrow \mathbb{R} \text{ s.t. } \int |f(x)| dx < \infty\}$. The operations $+$ and $\cdot$ are as above.
- (g) The space of all random variables $\mathcal{X} = \{X : X \text{ is a random variable}\}$. The operations $+$ and $\cdot$ are the usual addition and multiplication for random variables.
- (h) The space of all random variables with finite expectation: $\mathcal{L}_1(\mathcal{X}) = \{X : \mathbb{E}|X| < \infty\}$. $+$ and $\cdot$ are as above.
- (i) The space of all random variables with finite variance: $\mathcal{L}_2(\mathcal{X}) = \{X : \text{Var}(X) < \infty\}$. $+$ and $\cdot$ are as above.

Hint: A random variable X has finite variance if and only if $\mathbb{E}(X^2) < \infty$.

Note: Observe that the defining properties of a vector space has some consequences. In particular, the additive identity $\mathbf{0}$ is a unique element of the vector space – there cannot exist more than one element satisfying (A4). You can try to prove this by contradiction. Also, the additive inverse of a vector in (A5) is unique – there cannot exist more than one $\mathbf{y}$'s satisfying (A5) for a given $\mathbf{x}$. This is a consequence of $\mathbf{0}$ being a unique element. Moreover, the additive inverse is given by $-\mathbf{x} = (-1) \cdot \mathbf{x}$. This justifies the notation $-\mathbf{x}$.

Note: Sometimes a vector space is denoted by the triplet $(\mathcal{V}, +, \cdot)$ to explicitly specify the operations $+$ and $\cdot$. But, they can be ignored if they are clear from the context. It is very important to remember that the operations $+$ and $\cdot$ play a crucial role in determining whether a collection of objects is a vector space or not. The same collection $\mathcal{V}$ may be a vector space with $+$ and $\cdot$, but may not be a vector space with a different pair $(+', \cdot')$.

Exercise 2. A vector space can be defined in even more generality with $\mathbb{R}$ replaced by a **field**.

A collection of objects $\mathcal{F}$, equipped with two binary operations

$$+ : \mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F} \quad \text{and} \quad \cdot : \mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$$

is called a **field** if the following are satisfied.

(A1) $a, b \in \mathcal{F} \Rightarrow a + b \in \mathcal{F}$ (**closure** under $+$)

- (A2) $a + b = b + a \forall a, b \in \mathcal{F}$ (**commutativity** of $+$)
- (A3) $a + (b + c) = (a + b) + c \forall a, b, c \in \mathcal{F}$ (**associativity** of $+$)
- (A4) There exists $0 \in \mathcal{F}$ s.t. $a + 0 = a \forall a \in \mathcal{F}$. 0 is called the **additive identity**.
- (A5) For all $a \in \mathcal{F}$, there exists $b \in \mathcal{F}$ s.t. $a + b = 0$. Such a b is called the **additive inverse** of a and is denoted by $-a$.
- (M1) $a, b \in \mathcal{F} \Rightarrow a \cdot b \in \mathcal{F}$ (**closure** under $\cdot$)
- (M2) $a \cdot b = b \cdot a \forall a, b \in \mathcal{F}$ (**commutativity** of $\cdot$)
- (M3) $a \cdot (b \cdot c) = (a \cdot b) \cdot c \forall a, b, c \in \mathcal{F}$ (**associativity** of $\cdot$)
- (M4) There exists $1 \in \mathcal{F}$ s.t. $a \cdot 1 = a \forall a \in \mathcal{F}$. 1 is called the **multiplicative identity**.
- (M5) For all $a \in \mathcal{F}$, $a \neq 0$, there exists $b \in \mathcal{F}$ s.t. $a \cdot b = 1$. Such a b is called the **multiplicative inverse** of a and is denoted by a^{-1} .
- (D1) For all $a, b, c \in \mathcal{F}$, $a \cdot (b + c) = a \cdot b + a \cdot c$. ($\cdot$ is **distributive** over $+$)

Check that the following are fields.

- (a) The set of **real numbers** $\mathbb{R}$ equipped with the usual addition and multiplication.
- (b) The set of **rational numbers** $\mathbb{Q}$ equipped with the usual addition and multiplication.
- (c) The set of **complex numbers** $\mathbb{C}$ equipped with the addition and multiplication of complex numbers:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

- (d) $\mathcal{F} = \{0, 1\}$ with $+$ and $\cdot$ defined as

$$\begin{array}{ll} 0 + 0 = 0 & 0 \cdot 0 = 0 \\ 0 + 1 = 1 & 0 \cdot 1 = 0 \\ 1 + 0 = 1 & 1 \cdot 0 = 0 \\ 1 + 1 = 0 & 1 \cdot 1 = 1 \end{array}$$

Here, would it matter if we replace 0 and 1 by “cat” and “dog”, respectively? That is, $\mathcal{F} = \{\text{“cat”}, \text{“dog”}\}$ with $+$ and $\cdot$ defined as

$$\begin{array}{ll} \text{“cat”} + \text{“cat”} = \text{“cat”} & \text{“cat”} \cdot \text{“cat”} = \text{“cat”} \\ \text{“cat”} + \text{“dog”} = \text{“dog”} & \text{“cat”} \cdot \text{“dog”} = \text{“cat”} \\ \text{“dog”} + \text{“cat”} = \text{“dog”} & \text{“dog”} \cdot \text{“cat”} = \text{“cat”} \\ \text{“dog”} + \text{“dog”} = \text{“cat”} & \text{“dog”} \cdot \text{“dog”} = \text{“dog”} \end{array}$$

With the notion of a field in place, we now define a vector space in even more generality as follows.

Let $\mathcal{F}$ be a field and let $\mathcal{V}$ be a collection of objects, equipped with two operations

$$+ : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V} \quad \text{and} \quad \cdot : \mathcal{F} \times \mathcal{V} \rightarrow \mathcal{V}.$$

$\mathcal{V}$ with $+$ and $\cdot$ is called a **vector space over the field $\mathcal{F}$** if the following hold.

- (A1) $\mathbf{x}, \mathbf{y} \in \mathcal{V} \Rightarrow \mathbf{x} + \mathbf{y} \in \mathcal{V}$ ($\mathcal{V}$ is **closed** under $+$)

(A2) $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x} \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ (+ is **commutative**)

(A3) $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z} \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ (+ is **associative**)

(A4) There exists $\mathbf{0} \in \mathcal{V}$ s.t. $\mathbf{x} + \mathbf{0} = \mathbf{x} \forall \mathbf{x} \in \mathcal{V}$. Such an element $\mathbf{0}$ is called the **additive identity**.

(A5) For every $\mathbf{x} \in \mathcal{V}$, there exists a $\mathbf{y} \in \mathcal{V}$ such that $\mathbf{x} + \mathbf{y} = \mathbf{0}$. Such a $\mathbf{y}$ is called the **additive inverse** of $\mathbf{x}$ and is denoted by $-\mathbf{x}$.

(M1) $\mathbf{x} \in \mathcal{V} \Rightarrow a \cdot \mathbf{x} \in \mathcal{V} \forall a \in \mathcal{F}$ ($\mathcal{V}$ is **closed** under $\cdot$)

(M2) $a \cdot (b \cdot \mathbf{x}) = (a \cdot b) \cdot \mathbf{x} \forall a, b \in \mathcal{F}, \mathbf{x} \in \mathcal{V}$ ($\cdot$ is **commutative**)

Note. The $\cdot$'s on the left (i.e., $a \cdot (b \cdot \mathbf{x})$) are the binary operation $\cdot : \mathcal{V} \rightarrow \mathcal{V}$ defined on the space $\mathcal{V}$. For the $\cdot$'s on the right, the first one (i.e., $a \cdot b$) is the operation defined on the field $\mathcal{F}$, while the second one (i.e., $(a \cdot b) \cdot \mathbf{x}$) is the binary operation defined on $\mathcal{V}$.

(M3) $(a + b) \cdot \mathbf{x} = a \cdot \mathbf{x} + b \cdot \mathbf{x} \forall a, b \in \mathcal{F}, \mathbf{x} \in \mathcal{V}$ (+ is **distributive** over $\cdot$)

Note. The $+$ on the left (i.e., in $a + b$) is the operation defined on the field $\mathcal{F}$, whereas the $+$ on the right (i.e., in $a \cdot \mathbf{x} + b \cdot \mathbf{x}$) is the binary operation $+$ defined on $\mathcal{V}$.

(M4) $a \cdot (\mathbf{x} + \mathbf{y}) = a \cdot \mathbf{x} + a \cdot \mathbf{y} \forall a, b \in \mathcal{F}, \mathbf{x} \in \mathcal{V}$ ($\cdot$ is **distributive** over $+$).

(M5) $1 \cdot \mathbf{x} = \mathbf{x} \forall \mathbf{x} \in \mathcal{V}$. Here 1 is the multiplicative identity of $\mathcal{F}$.

As before, we call the elements of a vector space $\mathcal{V}$ (equipped with $+$ and $\cdot$, and over the field $\mathcal{F}$) as vectors. The operation $+$ is called **vector addition** and $\cdot$ is called **scalar multiplication**. The vector space is denoted by $(\mathcal{V}, +, \cdot, \mathcal{F})$ to explicitly denote the dependence on $+$, $\cdot$ and $\mathcal{F}$. The same collection $\mathcal{V}$ may or may not be a vector space when $+$, $\cdot$ and $\mathcal{F}$ change.

Also, two consequences of the definition are: the additive identity $\mathbf{0}$ is unique and is given by $0 \cdot \mathbf{x}$ for any $\mathbf{x} \in \mathcal{V}$, where 0 is the additive identity of $\mathcal{F}$; the additive inverse of any vector $\mathbf{x}$ is unique and is given by $-\mathbf{x} = (-1) \cdot \mathbf{x}$, where -1 is the additive inverse of 1 in $\mathcal{F}$.

(e) Check that $(\mathbb{C}^n, +, \cdot, \mathbb{C})$ is a vector space but $(\mathbb{R}^n, +, \cdot, \mathbb{C})$ is not a vector space.

(f) Check that the n -tuple of “cat” and “dog” is a vector space over the field $\mathcal{F}$ defined in (d) above, where the vector addition and scalar multiplication are defined elementwise, and the operations are as in (d) above. Here, by an n -tuple of “cat” and “dog” we mean a stacking of n elements, where each element is either “cat” or “dog”. Such a vector would be called a *vector of characters* or a *character vector*. This example shows that, in its full generality (and abstraction!), vectors need not have anything to do with numbers.

Note: Usually, for a vector space, the field $\mathcal{F}$ is taken to be either $\mathbb{R}$ or $\mathbb{C}$. A vector space over $\mathbb{R}$ is called a *real vector space*, while a vector space over $\mathbb{C}$ is called a *complex vector space*.

Exercise 3. Recall that a subset $\mathcal{S}$ of a vector space $\mathcal{V}$ is called a subspace if it satisfies the following:

$$\mathbf{x}, \mathbf{y} \in \mathcal{S} \Rightarrow \mathbf{x} + \mathbf{y} \in \mathcal{S} \quad \text{and} \quad \mathbf{x} \in \mathcal{S}, a \in \mathbb{R} \Rightarrow a \cdot \mathbf{x} \in \mathcal{S}.$$

Here, $+$ and $\cdot$ are the vector addition and scalar multiplication operations from the vector space $\mathcal{V}$.

(a) Verify that since $\mathcal{S}$ is a subset of a vector space, the two conditions above are enough to ensure that $\mathcal{S}$ itself is a vector space (when equipped with $+$ and $\cdot$ from its superset $\mathcal{V}$).

(b) Use this to verify that, in the previous exercise, $\mathcal{F}_c, \mathcal{F}_1$ are subspaces of $\mathcal{F}$, and $\mathcal{L}_1(\mathcal{X}), \mathcal{L}_2(\mathcal{X})$ are subspaces of $\mathcal{X}$. Moreover, $\mathcal{L}_2(\mathcal{X})$ is a subspace of $\mathcal{L}_1(\mathcal{X})$. Hence deduce that $\mathcal{F}_c, \mathcal{F}_1, \mathcal{L}_1(\mathcal{X}), \mathcal{L}_2(\mathcal{X})$ are vector spaces. This verification is much simpler than in the previous exercise.